

Evaluating Experimental Designs for Short Video Studies

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OVERVIEW

Existing studies focused on optimizing the recommendation and preloading mechanisms to enhance the user experience using conventional and learning based approaches. However, existing studies use different experimental designs to evaluate their mechanisms, which could potentially lead to less reliable results in various environments. Hence, in this work we explore various experimental designs and evaluate the efficiency of existing mechanisms under various experimental designs. We provide guidelines to make better experimental design choices.

METHOD

Short Video Mechanisms

Joint Preload

- ❑ Bandwidth estimation
- ❑ Chunk level probability of stay

Network Aware

- ❑ Bandwidth estimation
- ❑ Decides buffer length and preload number of videos based on Bandwidth estimation

Video Traces

- ❑ 7 videos, durations 6 to 125 seconds
- ❑ Bitrate: 0.73 Mbps, 1.17 Mbps, 1.81 Mbps
- ❑ Chunk duration 1 second

Network Traces

- ❑ High, low, medium, mixed categories
- ❑ Minimum 0.2 Mbps
- ❑ Maximum 14.43 Mbps

Dataset

User Behavior Traces

- ❑ Actual: 384.5 secs
- ❑ Step 1 increase: 550.68 secs
- ❑ Step 2 increase: 690.1 secs
- ❑ Step 1 decrease: 207.57 secs
- ❑ Step 2 decrease : 83.14 secs

EXPERIMENTAL SETUP

- ❑ Fixed number of videos in a session – 35 videos in a session
- ❑ Fixed session duration including stall durations – 600 seconds session duration
- ❑ Fixed session duration without stall durations – 600 seconds session duration

QUALITY OF EXPERIENCE (QoE) FORMULA

$$QoE = w_1 \times \text{bitrate} - w_2 \times \text{rebuffer duration} - w_3 \times \text{bitrate variation}$$

$$\text{data waste} = \text{total bitrate of downloaded duration of the video} - \text{total bitrate of played duration of the video}$$

$$QoE_{\text{waste}} = QoE - w_4 \times \text{data waste}$$

$$w_1 = 1, \quad w_2 = 1.85, \quad w_3 = 1, \quad w_4 = 0.5$$

RESULTS

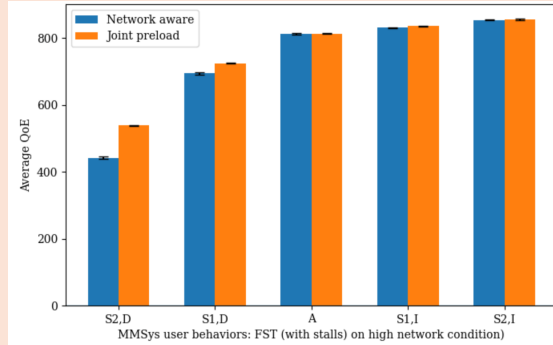


Figure 1: QoE components performance

Figure 1 shows that, QoE score favors the high watch duration. More the watch duration better the QoE score despite of high re-buffering duration.

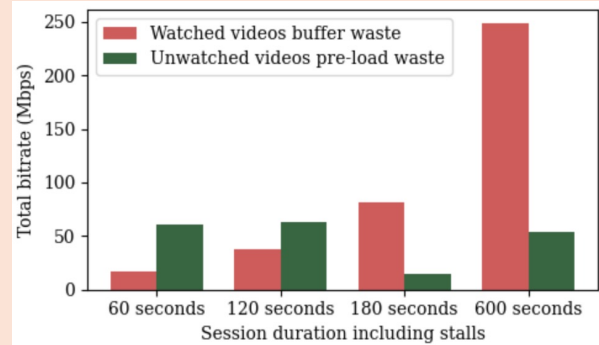


Figure 2: Preload waste

Figure 2 shows that, increase in preload waste compared to watched video buffer waste in smaller session duration. However, as session duration increases preload waste reduce.

NEXT STEPS

Develop a standardized experimental design for evaluating the short video mechanisms. Re-formulate the QoE formula that captures the session dynamics in the short video platform. Explore more on the duration, bitrates of the chunks and the bitrate variation.